# Assignment 3: Reference and Rhetorical Purpose Questions

## The Smartphone Revolution

**1** Smartphones have rapidly increased their dominance in the mobile phone market in recent years, accounting for more than half of all mobile phones sold globally. These multifunctional devices have revolutionized the way in which users communicate, correspond, and interact with the world, and others, around them. Since smartphones have combined the tasks of previously separate devices into one exceptionally efficient product, their appeal to consumers cannot be overestimated. Prior to the modern development of the smartphone, people required a combination of machines, such as computers, televisions, PDAs (Personal Digital Assistants), and cameras to accomplish all of the functions that are currently handled with a single piece of technology.

**2** While smartphones have only been in existence since the 1990s, the original concept was introduced in 1973 by Theodore Paraskevakos. In that year, the innovative entrepreneur obtained a patent for his idea of uniting the functions of data processing activities, intelligent applications, and visual displays with those of a telephone. At that time, Paraskevakos made note of such tasks as banking and bill paying in his outline; two commonplace activities many users today perform with their smartphones. Paraskevakos may have come up with his concept of an advanced phone as a result of his earlier success with transmitting electronic data through telephone lines, a process that became the foundation for the "Caller ID (Identification)" function available on virtually all contemporary phones. Since beginning his engineering work in 1968, Paraskevakos has obtained over 20 patents worldwide which are based on this technology.

**3** Despite the fact that it took almost 20 years to bring the first smartphone to market, once it appeared, subsequent generations of enhanced versions have emerged with increasing frequency, offering consumers a variety of brands from which to choose. In 1992, the IBM Corporation demonstrated a prototype of a phone that incorporated PDA capabilities with traditional phone functions. A couple of years later, an improved version was put on the market for the public. By this time, several competitors were working on their own adaptations of what would later be called a "smartphone," and by the end of the first decade of the twenty-first century, the popularity of the smartphone approached that of its predecessor, the "feature" phone.

**4** Though mobile phones are divided into three categories—those that provide only basic phone and text services, feature phones, and smartphones—the line between a feature phone and a smartphone is still somewhat blurred. In general, a feature phone differs from a smartphone in that it provides access to less functions than the smartphone does, although it does have a variety of functions over and above those of a basic mobile phone. One of the challenges of distinguishing between feature phones and smartphones is that, with the rapid evolution of successive generations of phones, the features exclusive to yesterday's smartphones are often found on today's feature phones. So while modern feature phones may still be behind in terms of the capabilities of their smartphone contemporaries, they may in fact be more advanced than smartphones of just a few years ago. For example, in the early part of the first decade of the twenty-first century, functions such as GPS (Global Positioning System) and Wi-Fi (Wireless Fidelity) access belonged solely to smartphones, whereas, by the end of that decade, many feature phones had evolved to include these functions.

**5** One characteristic that most people agree differentiates smartphones from feature phones

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is the use of “apps,” application programs designed specifically for the individual operating systems of smartphones. These apps are programs that can be downloaded directly onto the smartphone, or downloaded through a website. Apps have become one of the most popular features of smartphones in recent years, and cover a wide range of fields. Some apps, such as word processing or spreadsheet programs, help users complete tasks. Other apps provide information, such as weather or location. Still others are a source of entertainment, allowing for books and music to be downloaded, or for games to be played on the smartphone screen. For example, Angry Birds, one of the most popular apps on the market when it was created, was a game first devised for the Apple
iPhone. Its enormous fame led to versions being devised for other smartphone operating systems, and then for other devices, such as computers and gaming consoles.

1. The word “their” in paragraph 1 refers to
   A. smartphones
   B. tasks
   C. separate devices
   D. consumers

2. The word “those” in paragraph 2 refers to
   A. intelligent applications
   B. functions
   C. data processing activities
   A. displays

3. The word “this technology” in paragraph 2 refers to
   A. contemporary phones
   B. telephone lines
   C. electronic data
   D. Caller ID

4. The phrase “an improved version” in paragraph 3
refers to
   A. a variety
   B. a phone
   C. a PDA
   D. a prototype

5. The word “its” in paragraph 3 refers to
   A. the first decade
   B. the public
   C. the smartphone
   D. the feature phone

6. The word “they” in paragraph 4 refers to
   A. feature phones
   B. capabilities
   C. contemporaries
   D. functions

7. The word “others” in paragraph 5 refers to
   A. programs
   B. tasks
   C. apps
   D. books

8. The word “Its” in paragraph 5 refers to
   A. Angry Birds
   B. the market
   C. the Apple iPhone
   D. a game

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## Xerography

**1** One more familiar use of electrochemistry that has made its way into the mainstream is xerography, a process for replicating documents that is dependent on photoconductive materials. A photoconductive material is an insulator in the dark but becomes a conductor when exposed to bright light. When a photocopy is being made, an image of a document is projected onto the surface of a rotating drum, and bright light causes the photoconductive material on the surface of the drum to become conductive.

**2** As a result of the conductivity, the drum loses its charge in the lighted areas, and toner (small grains to which dry ink adheres) attaches itself only to the darker parts of the image. The grains are then carried to a sheet of paper and fused with heat. When a laser printer is used, the image is projected by means of a laser beam, which creates a brighter light and a greater contrast between lighter and darker areas and therefore results in sharper printed images.

**3** Xerography has gone through a series of innovations since its invention in the late 1930s by Chester Carlson. One of the most notable alterations was to the name of the process. When Carlson first developed the technique, he called it "electrophotography" since it used both photography and electrostatic printing in the procedure. Later the name was changed to "xerography" in recognition of the fact that the process reproduces documents without the use of liquid chemicals, instead using a powdered toner to replicate the images.

**4** A more significant change was made to the instruments that actually created the copies. Initially, the procedure took several steps to complete, and necessitated the use of flat plates that were manipulated by hand through the various copying stages. From the beginning, Carlson and others involved with the process realized how inconvenient and time-consuming it was to make a copy, and they worked continuously on ideas to make the process faster and more efficient. Eighteen years after the original machine was introduced to the public, they devised a suitable solution to the flat plates: a cylindrical rotating surface that allowed the process to be entirely automatic, except for a push of the "start" button.

**5** The revolutionary invention of the rotating drum meant the copier was now viable as a commercial product. The first commercial automatic copier, the Xerox 914, came onto the market in 1960. This first generation of copiers was cumbersome by today's standards; not many households would be able to dedicate the necessary space it would take to have one of these early machines. Fortunately, continuing innovations in both the xerographic process and other types of technology have greatly decreased the size of machines capable of making copies. Today, the vast majority of copy machines, as well as many laser and LED printers, make use of Carlson's ingenious idea.

**6** Photocopying through xerography involves several steps. First, an electrostatic charge is evenly spread over the surface of the rotating drum, or cylinder. The distributed charge is positive or negative depending on what type of copy is being made and what type of copier is being used. Standard copiers generally distribute a positive charge, while digital copiers use a negative charge. Next, the document being duplicated is exposed to light by flash lamps. At the same time, a combination of lenses and mirrors projects the original image through a lens, so that it is projected onto and synchronized with the rotating drum.

**7** The third and fourth steps of the process concern the development and transfer of the image. In the development stage, a form of static electricity propels toner powder to coat the image that was projected onto the drum in the previous step. Then, in the transfer step, the

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toner from the drum, now in the form of the image, is transferred onto paper as the paper passes through the copying mechanism. The transfer is achieved through a blend of pressure on the paper and electrostatic attraction of the toner powder to the paper.

**8** The remaining steps of the copying process finalize the image onto the paper. After the image has been transferred, the next step is to neutralize the electric charge on the paper and separate the paper from the drum surface. After that, the toner is permanently bonded to the paper using heat or a radiant fusing process, both of which ensure that the toner particles are permanently affixed to the paper. Finally, any remaining toner on the drum is cleaned off through a process that typically includes some type of suction to remove the particles. Usually, this toner is carried to a container in the machine for later disposal, though some machines will recycle the toner for use in subsequent copies.

1. The author begins the first paragraph with "One more familiar use of electrochemistry" in order to
   A. explain that xerography is one of the less familiar uses of electrochemistry
   B. make it clear that electrochemistry requires photoconductive materials
   C. show that xerography is the only known use for electrochemistry
   D. indicate that other less familiar uses have already been discussed

2. Why does the author explain that "A photoconductive material is an insulator in the dark but becomes a conductor when exposed to bright light"?
   A. It gives an explanation of a property that is necessary for xerography.
   B. It indicates that bright light is required for insulation to take place.
   C. It gives one example of a successful xerographic process.
   D. It explains the role of insulation in xerography.

3. The author mentions "smail grains to which dry ink adheres" in order to
   A. provide information that contradicts the previous statement
   B. provide another example of conductivity
   C. provide further detail information about toner
   D. provide an alternate explanation for the effectiveness of toner

4. Why does the author mention "a laser printer" in the passage?
   A. It is an alternative to xerography.
   B. It is a way of duplicating without using electrochemistry.
   C. It is a second example of xerography.
   D. It is a less effective type of xerography than is a photocopier.

5. Why does the author include the phrase "except for a push of the 'start' button"?
   A. to explain the one step of the process that Carlson had no control over
   B. to indicate that it is not necessary to push a button to begin the copying process
   C. to emphasize that improvements to the copying process reduced the amount of work people had to do
   D. to show that Carlson had thought of almost every step necessary in the copying process

6. The author mentions that "not many households would be able to dedicate the necessary space" for a copier in order to
   A. explain that copiers remain exclusively a product for businesses
   B. emphasize how large the first automatic copiers were
   C. illustrate the dislike most households had for copiers when they were first introduced
   D. explain how households have increased in size since 1960

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7. Why is “a blend of pressure on the paper and electrostatic attraction of the toner powder” mentioned?

A. to summarize the final step of the copying process
B. to illustrate how toner is propelled in the development stage
C. to explain how the drum is coated with powder
D. to explain how the image is transferred onto paper

8. Why does the author discuss suction in paragraph 8?

A. to provide an explanation of how the last part of the copying process is achieved
B. to illustrate the neutralization process
C. to give an example of how toner is recycled
D. to explain how the toner adheres to the paper